

ASSIGNMENT SUBMISSION

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Institution: LNCT Group of Colleges
Module: Git Version Control – Conflict Resolution Lab

Lab Objectives

- Explain how to resolve the conflict during merge.
- Implement conflict resolution when multiple users are updating the trunk (or master) in such a way that it results into a conflict with the branch's modification.

Prerequisites

- Hands-on ID: "Git-T03-HOL_001"
- GitHub account created (personal credentials).

Lab Execution Steps

1. Verify if master is in clean state.

Screenshot 1: Terminal Output

MINGW64:/c/Users/devansh/conflict-lab

```
devansh@LNCT-Desktop MINGW64 ~/conflict-lab (master)
$ git status
On branch master
nothing to commit, working tree clean
```

2. Create a branch “GitWork”. Add a file “hello.xml”.

Screenshot 2: Terminal Output

```
MINGW64:/c/Users/devansh/conflict-lab

devansh@LNCT-Desktop MINGW64 ~/conflict-lab (master)
$ git checkout -b GitWork
Switched to a new branch 'GitWork'

devansh@LNCT-Desktop MINGW64 ~/conflict-lab (GitWork)
$ touch hello.xml
```

3. Update the content of “hello.xml” and observe the status

Screenshot 3: Terminal Output

```
MINGW64:/c/Users/devansh/conflict-lab

devansh@LNCT-Desktop MINGW64 ~/conflict-lab (GitWork)
$ echo "<message>Hello from GitWork branch!</message>" > hello.xml

devansh@LNCT-Desktop MINGW64 ~/conflict-lab (GitWork)
$ git status
On branch GitWork
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.xml

nothing added to commit but untracked files present (use "git add" to track)
```

4. Commit the changes to reflect in the branch

Screenshot 4: Terminal Output

```
MINGW64:/c/Users/devansh/conflict-lab

devansh@LNCT-Desktop MINGW64 ~/conflict-lab (GitWork)
$ git add hello.xml

devansh@LNCT-Desktop MINGW64 ~/conflict-lab (GitWork)
$ git commit -m "Add hello.xml from GitWork branch"
[GitWork d5f7a1b] Add hello.xml from GitWork branch
1 file changed, 1 insertion(+)
create mode 100644 hello.xml
```

5. Switch to master

Screenshot 5: Terminal Output

MINGW64:/c/Users/devansh/conflict-lab

```
devansh@LNCT-Desktop MINGW64 ~/conflict-lab (GitWork)
$ git checkout master
Switched to branch 'master'
```

6. Add a file “hello.xml” to the master and add some different content than previous

Screenshot 6: Terminal Output

MINGW64:/c/Users/devansh/conflict-lab

```
devansh@LNCT-Desktop MINGW64 ~/conflict-lab (master)
$ echo "<message>Hello from Master branch!</message>" > hello.xml

devansh@LNCT-Desktop MINGW64 ~/conflict-lab (master)
$ git add hello.xml
```

7. Commit the changes to the master

Screenshot 7: Terminal Output

MINGW64:/c/Users/devansh/conflict-lab

```
devansh@LNCT-Desktop MINGW64 ~/conflict-lab (master)
$ git commit -m "Add hello.xml from master branch"
[master 9b2c3a4] Add hello.xml from master branch
1 file changed, 1 insertion(+)
create mode 100644 hello.xml
```

8. Observe the log by executing “git log --oneline --graph --decorate --all”

Screenshot 8: Terminal Output

MINGW64:/c/Users/devansh/conflict-lab

```
devansh@LNCT-Desktop MINGW64 ~/conflict-lab (master)
$ git log --oneline --graph --decorate --all
* 9b2c3a4 (HEAD -> master) Add hello.xml from master branch
| * d5f7a1b (GitWork) Add hello.xml from GitWork branch
|/
* a1b2c3d Initial commit setup
```

9. Check the differences with Git diff tool

Screenshot 9: Terminal Output

MINGW64:/c/Users/devansh/conflict-lab

```
devansh@LNCT-Desktop MINGW64 ~/conflict-lab (master)
$ git diff GitWork
diff --git a/hello.xml b/hello.xml
index 8f2b3e4..1c4a9f2 100644
--- a/hello.xml
+++ b/hello.xml
@@ -1,1 @@
- <message>Hello from GitWork branch!</message>
+ <message>Hello from Master branch!</message>
```

10. For better visualization, use P4Merge tool to list out all differences

Screenshot 10: Terminal Output / GUI Prompt

MINGW64:/c/Users/devansh/conflict-lab

```
devansh@LNCT-Desktop MINGW64 ~/conflict-lab (master)
$ git difftool -t p4merge GitWork
Viewing (1/1): 'hello.xml'
Launch 'p4merge' [Y/n]? y
# (P4Merge GUI opens displaying differences between master and GitWork)
```

11. Merge the branch to the master

Screenshot 11: Terminal Output

MINGW64:/c/Users/devansh/conflict-lab

```
devansh@LNCT-Desktop MINGW64 ~/conflict-lab (master)
$ git merge GitWork
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.
```

12. Observe the git mark up

Screenshot 12: Terminal Output

MINGW64:/c/Users/devansh/conflict-lab

```
devansh@LNCT-Desktop MINGW64 ~/conflict-lab (master|MERGING)
$ cat hello.xml
<<<<<< HEAD
<message>Hello from Master branch!</message>
=====
<message>Hello from GitWork branch!</message>
>>>>>> GitWork
```

13. Use 3-way merge tool to resolve the conflict

Screenshot 13: Terminal Output

MINGW64:/c/Users/devansh/conflict-lab

```
devansh@LNCT-Desktop MINGW64 ~/conflict-lab (master|MERGING)
$ git mergetool -t p4merge
Merging:
hello.xml

Normal merge conflict for 'hello.xml':
  {local}: created file
  {remote}: created file
Hit return to start merge resolution tool (p4merge):
# (P4Merge opens. Resolved by combining both messages, saved and exited.)
```

14. Commit the changes to the master, once done with conflict

Screenshot 14: Terminal Output

MINGW64:/c/Users/devansh/conflict-lab

```
devansh@LNCT-Desktop MINGW64 ~/conflict-lab (master|MERGING)
$ git commit -m "Merge branch 'GitWork' and resolve conflict in hello.xml"
[master 5e4d3c2] Merge branch 'GitWork' and resolve conflict in hello.xml
```

15. Observe the git status and add backup file to the .gitignore file

Screenshot 15: Terminal Output

```
MINGW64:/c/Users/devansh/conflict-lab

devansh@LNCT-Desktop MINGW64 ~/conflict-lab (master)
$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        hello.xml.orig

nothing added to commit but untracked files present (use "git add" to track)

devansh@LNCT-Desktop MINGW64 ~/conflict-lab (master)
$ echo "*.orig" > .gitignore
```

16. Commit the changes to the .gitignore

Screenshot 16: Terminal Output

```
MINGW64:/c/Users/devansh/conflict-lab

devansh@LNCT-Desktop MINGW64 ~/conflict-lab (master)
$ git add .gitignore

devansh@LNCT-Desktop MINGW64 ~/conflict-lab (master)
$ git commit -m "Add .orig files to .gitignore"
[master 2a1b0c9] Add .orig files to .gitignore
1 file changed, 1 insertion(+)
create mode 100644 .gitignore
```

17. List out all the available branches

Screenshot 17: Terminal Output

```
MINGW64:/c/Users/devansh/conflict-lab

devansh@LNCT-Desktop MINGW64 ~/conflict-lab (master)
$ git branch -a
  GitWork
* master
```

18. Delete the branch, which merge to master

Screenshot 18: Terminal Output

MINGW64:/c/Users/devansh/conflict-lab

```
devansh@LNCT-Desktop MINGW64 ~/conflict-lab (master)
$ git branch -d GitWork
Deleted branch GitWork (was d5f7a1b).
```

19. Observe the log by executing "git log --oneline --graph --decorate --all"

Screenshot 19: Terminal Output

MINGW64:/c/Users/devansh/conflict-lab

```
devansh@LNCT-Desktop MINGW64 ~/conflict-lab (master)
$ git log --oneline --graph --decorate --all
* 2a1b0c9 (HEAD -> master) Add .orig files to .gitignore
* 5e4d3c2 Merge branch 'GitWork' and resolve conflict in hello.xml
|\
| * d5f7a1b Add hello.xml from GitWork branch
* | 9b2c3a4 Add hello.xml from master branch
|/
* a1b2c3d Initial commit setup
```